

# ■ CONVECTION CURRENTS — COLOUR IN WATER

Detailed Lesson Plan | Science Experiment for Kids

| ■ Duration | ■ Type              | ■ Location       |
|------------|---------------------|------------------|
| 60–75 mins | Hands-on Experiment | Classroom / Home |

## ■ Lesson Overview

This engaging, hands-on science lesson introduces young learners to the concept of **convection currents** through a colourful water experiment. Students observe how heat causes water to move in circular loops — hot water rising and cold water sinking — making the invisible visible through food colouring. Indian daily-life examples (sambar, agarbatti, sea breeze) make the concept relatable and memorable.

## ■ Learning Objectives

By the end of this lesson, students will be able to:

- Define convection current in their own words.
- Explain why hot water rises and cold water sinks.
- Distinguish between conduction and convection with examples.
- Identify real-life examples of convection currents in India.
- Make and test scientific predictions (predict & observe).

## ■ Materials Required

| Item                                 | Quantity | Notes                  |
|--------------------------------------|----------|------------------------|
| Clear glass or transparent container | 2        | Tall glasses work best |

|                                  |               |                             |
|----------------------------------|---------------|-----------------------------|
| Hot water (warm, not boiling)    | 1 cup         | Adult to prepare            |
| Cold water / ice water           | 1 cup         | Use blue dye                |
| Food colouring / turmeric powder | 2 colours     | Yellow/orange + blue        |
| Dropper or spoon                 | 1–2           | For adding colour carefully |
| Tray / newspaper                 | 1             | To catch spills             |
| Worksheet / observation sheet    | 1 per student | Printed beforehand          |

## ■ Lesson Plan (Step-by-Step)

### ■ Phase 1: Hook & Engagement

■ 10 minutes

#### Open with a question:

Ask students: "Have you ever seen steam rise from a hot cup of chai? Why does it rise UP and not go DOWN?" Allow 2–3 student responses.

#### Story starter:

Introduce Kabir (from the lesson PDF) — a curious boy who drops turmeric into hot water and sees the yellow colour swirl upward. Ask: "Why did the colour go UP?"

#### Connect to prior knowledge:

Ask if anyone has seen ink or food colouring spread in water. What did it look like?

### ■ Phase 2: Concept Introduction

■ 10 minutes

#### Explain with analogy:

Hot water is like a helium balloon — it becomes lighter and floats UP. Cold water is like a stone — it becomes heavier and sinks DOWN.

#### Introduce the term:

Write CONVECTION CURRENT on the board. Say: "When hot parts go up and cool parts come down in a continuous loop, we call it a convection current."

#### Golden Rule:

Hot Rises ↑ — Cold Sinks ↓ — Creates a LOOP.

### ■ Phase 3: The Experiment

■ 20 minutes

#### Step 1 — Set up:

Fill one glass with cold water. Add 2 drops of blue food colouring. Fill another with warm water. Add 2 drops of yellow/orange colouring.

#### Step 2 — Predict:

Before mixing, ask students to predict: "If we gently pour warm coloured water into the cold glass, what will happen?" Record predictions on the board.

**Step 3 — Observe:**

Slowly pour warm orange water into cold blue water. Ask students to watch where each colour travels. Hot orange rises; cold blue sinks.

**Step 4 — Discuss:**

"Was your prediction correct? What did you see? Why did the colours separate?"

**■ Phase 4: Deepen Understanding****■ 10 minutes****Conduction vs Convection:**

Draw two columns on the board. Conduction = heat moves by TOUCHING (metal spoon in dal). Convection = heat moves by MOVING (water rising in a pot).

**Memory trick:**

TOUCH = conDUCTION. MOVE = conVECTION. V for Velocity/Movement!

**Real-life Indian examples:**

Sea breeze in Mumbai (land heats faster than sea). Sambar bubbling on a slow flame. Agarbatti smoke rising in a temple.

**■ Phase 5: Activity & Assessment****■ 10 minutes****Mini Quiz (oral or written):**

Q1: What happens to hot water? Q2: What is a convection current? Q3: What is heat?

**Drawing activity:**

Students draw a labelled diagram in their notebooks: a glass of water showing hot water (arrows up, orange) and cold water (arrows down, blue).

**Exit ticket:**

"Name ONE place in your home where you see convection currents every day."

**■ Phase 6: Wrap-Up & Recap****■ 5 minutes****Recap points:**

1) Heat makes water loop — called convection current. 2) Hot rises, cold sinks. 3) Colour in water makes this visible. 4) Convection ≠ Conduction. 5) We see this in daily life!

**Encourage exploration:**

"Try this at home tonight with a parent's help! What other liquids can you try?"

## ■ Differentiation Strategies

| ■ For Struggling Learners                                                                                                                                                                            | ■ For Advanced Learners                                                                                                                                                                                                           | ■ For All Learners                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Use only two colours (blue/red)</li> <li>• Provide sentence starters</li> <li>• Pair with a stronger student</li> <li>• Use visual diagram cards</li> </ul> | <ul style="list-style-type: none"> <li>• Ask: where else does this happen? (Earth's mantle, ocean currents)</li> <li>• Introduce density as the science reason</li> <li>• Challenge: compare salt water vs fresh water</li> </ul> | <ul style="list-style-type: none"> <li>• Group discussion after experiment</li> <li>• Peer teaching — explain to a partner</li> <li>• Relate to local examples</li> <li>• Use think-pair-share routine</li> </ul> |

## ■ Assessment Rubric

| Criteria                        | Excellent (4)                                        | Good (3)                            | Developing (2)                     | Needs Support (1)     |
|---------------------------------|------------------------------------------------------|-------------------------------------|------------------------------------|-----------------------|
| <b>Concept Understanding</b>    | Clearly explains convection with correct terminology | Explains with correct terminology   | Partially explains, some confusion | Cannot explain yet    |
| <b>Observation Skills</b>       | Accurately describes color and movement              | Describes movement with details     | Notes some movement                | Lacks detail          |
| <b>Prediction &amp; Testing</b> | Makes logical prediction                             | Makes prediction, partially correct | Makes a guess; difficulty          | No prediction made    |
| <b>Real-life Connections</b>    | Names 2+ Indian examples                             | Names 1 example with explanation    | Names 1 example but cannot explain | Cannot explain        |
| <b>Diagram / Drawing</b>        | Labelled diagram with arrows and colors              | Diagram with arrows and some labels | Basic labeling, few labels         | Incomplete or missing |

## ■ Student Observation Worksheet

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

### ■ Before the Experiment — My Prediction

I think when we pour warm orange water into cold blue water, the orange will go...

☐ Up to the top    ☐ Down to the bottom    ☐ Mix evenly

Because I think:

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### ■ During the Experiment — What I Observed

The orange coloured water moved: ☐ UP    ☐ DOWN    ☐ Stayed in the middle

The blue coloured water moved: ☐ UP    ☐ DOWN    ☐ Stayed in the middle

Describe in your own words what you saw:

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### ■ After the Experiment — What I Learned

A convection current is:

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Hot water rises because:

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Cold water sinks because:

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One Indian example of convection:

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⇒ **Draw It! — Label your diagram with arrows showing hot and cold water movement.**

*(Draw your glass of water below, showing hot water rising and cold water sinking with coloured arrows)*

↑ (Hot) \_\_\_\_\_ colour      ↓ (Cold) \_\_\_\_\_ colour      ■ Convection  
Loop

## Teacher Notes & Safety

### Safety Reminders

- Always use warm (not boiling) water to avoid burns.
- Adults/teachers must pour hot water.
- Place tray under glasses to manage spills.
- Supervise at all times during experiment.
- Brief students on safety BEFORE starting.

### Pedagogical Tips

- Use turmeric if food dye unavailable — very visible!
- Let students make predictions before seeing results.
- Encourage 'I wonder...' questions throughout.
- Revisit Indian examples to connect science to culture.
- Extend: Discuss ocean currents and weather patterns.

## Curriculum Links & Extensions

| Subject                 | Connection                                     | Extension Activity                              |
|-------------------------|------------------------------------------------|-------------------------------------------------|
| Science (EVS / Physics) | Heat transfer, states of matter, density       | Research ocean thermal energy conversion (OTEC) |
| Geography               | Sea breeze, land breeze, monsoon wind patterns | Map Indian coastlines; discuss sea breezes      |

|                                  |                                                    |                                              |
|----------------------------------|----------------------------------------------------|----------------------------------------------|
| <b>Mathematics</b>               | Temperature scale (degrees C),<br>measuring volume | Graph temperature vs. time as water<br>cools |
| <b>Environmental<br/>Science</b> | Global warming, ocean currents,<br>climate         | Discuss how convection affects weather       |
| <b>Language Arts</b>             | Scientific vocabulary, describing<br>observations  | Write a short paragraph: The Water<br>Dance  |

■ **Homework:** With a parent's help, repeat the experiment at home. Try hot chai-coloured water vs cold water. Observe for 2 minutes. Write 3 sentences about what you saw in your science diary.